Laboratory Activity 6

Use Case Diagram Development ? CategorAIze ? Parts 1?3

Part 1 ? Project scope

CategorAIze helps email account owners organize and prioritize messages, understand email content, and act on tasks and deadlines. It supports English and Filipino/Taglish analysis, user-defined categories, prediction corrections, email search, reply preparation, inbox cleanup, and Google Calendar integration.

System boundary: the CategorAIze web application, background processing, database, and internal NLP service. Gmail and Google Calendar are external services. This model follows documented target workflows; it does not claim every planned capability is already implemented or instructor-approved.

In scope: account connection, synchronization, category management, analysis review, prediction feedback, search, task/deadline review, reply drafts, calendar availability, explicitly approved sending, inbox cleanup, and confirmed calendar export.

Out of scope: deployment administration, model-training operations, billing, and inherited features unrelated to core workflows. No administrator role is assumed because the selected workflows define no separate administrator-facing service.

Actor inventory

Actor	Description and main interaction
A1 ? Email Account Owner	Primary human actor who connects and manages their mailbox. Connects an account, reviews and organizes email, corrects predictions, searches messages, manages tasks, prepares replies, approves sending, cleans the inbox, and exports calendar events.
A2 ? Gmail Service	External email and OAuth service. Authorizes mailbox access, supplies messages and updates, stores drafts, sends approved replies, and applies requested mailbox changes.
A3 ? Google Calendar Service	External calendar service linked with the owner's consent. Supplies availability and creates events after the owner confirms an export.

The NLP service and database are internal components. Email senders reach CategorAIze through Gmail and do not directly operate the selected services.

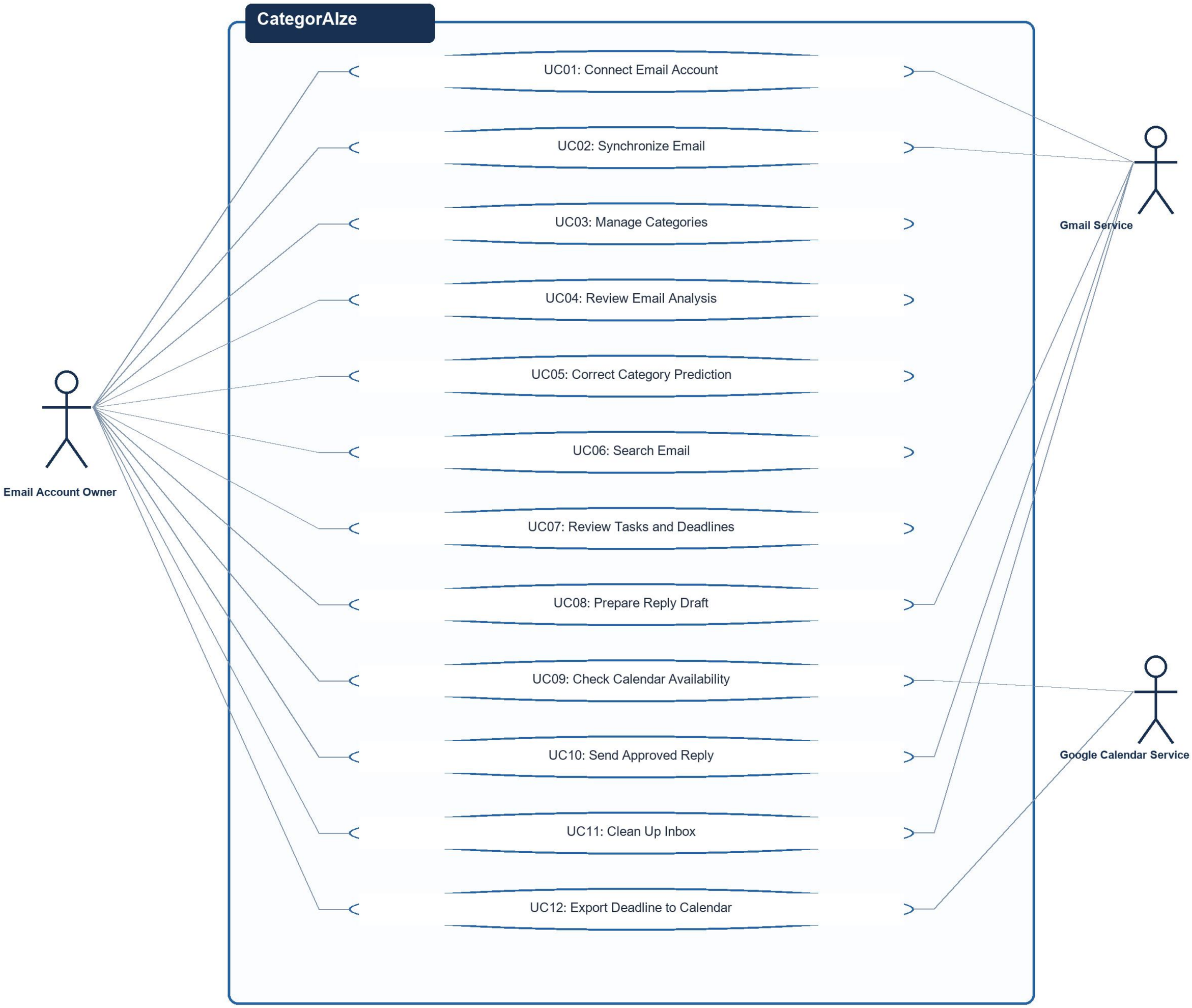
Part 2: Use Case Identification

Actor, use case, and purpose inventory

Actor	Use Case	Purpose
Email Account Owner	Connect Email Account	Allows the owner to sign in and authorize CategorAIze to access Gmail.
Gmail Service	Connect Email Account	Provides OAuth authorization and mailbox access.
Email Account Owner	Synchronize Email	Allows the owner to retrieve new and updated messages for processing and review.
Gmail Service	Synchronize Email	Supplies new and updated messages from the connected mailbox.
Email Account Owner	Manage Categories	Allows the owner to create, edit, activate, or remove categories and set classification preferences.
Email Account Owner	Review Email Analysis	Allows the owner to inspect category predictions, priority explanations, summaries, actions, and deadlines when available.
Email Account Owner	Correct Category Prediction	Allows the owner to submit an explicit correction using an existing category to improve personalized predictions.
Email Account Owner	Search Email	Allows the owner to find relevant messages using semantic search and available filters.
Email Account Owner	Review Tasks and Deadlines	Allows the owner to review extracted actions, approaching deadlines, and dates that need attention.
Email Account Owner	Prepare Reply Draft	Allows the owner to generate and edit a contextual reply before saving it as a Gmail draft.
Gmail Service	Prepare Reply Draft	Stores the reply in Gmail Drafts when requested.
Email Account Owner	Check Calendar Availability	Allows the owner to consult calendar availability for scheduling or a reply.
Google Calendar Service	Check Calendar Availability	Supplies the connected calendar's availability information.
Email Account Owner	Send Approved Reply	Allows the owner to send a reviewed reply after explicitly confirming it.
Gmail Service	Send Approved Reply	Sends the reply after receiving the owner's confirmed request.
Email Account Owner	Clean Up Inbox	Allows the owner to archive selected messages, unsubscribe from unwanted mail, or configure cold-outreach blocking.
Gmail Service	Clean Up Inbox	Applies requested mailbox changes and supports unsubscribe actions.
Email Account Owner	Export Deadline to Calendar	Allows the owner to confirm a reviewed deadline before creating a calendar event.
Google Calendar Service	Export Deadline to Calendar	Creates the event after the owner confirms its details.

Part 3: UML Use Case Diagram

CategorAIze - system boundary, external actors, and associations



Solid associations show actor participation; they do not indicate sequence or data flow.